

Generalized Linear Models

STA 310: Homework 5

1. The probability density function (PDF) of the Pareto Type I distribution is given by:

$$f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}} \mathbf{1}_{\{x \geq x_m\}}$$

where:

- $\mathbf{1}_{\{x \geq x_m\}}$ is the indicator function, which is 1 if $x \geq x_m$ and 0 otherwise.
- $\alpha > 0$ is the shape parameter.
- $x_m > 0$ is the scale parameter (minimum possible value) of the data.

Is this a member of the exponential family (in canonical form), where both parameters are unknown? If so, identify the components in canonical form. If not, explain why not.

$$f(x \mid \alpha, x_m) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}} \mathbf{1}_{\{x \geq x_m\}}, \quad \alpha > 0, x_m > 0.$$

Take logs:

$$\log f(x \mid \alpha, x_m) = \log \alpha + \alpha \log x_m - (\alpha + 1) \log x + \log \mathbf{1}_{\{x \geq x_m\}}.$$

If a distribution is in the (regular) exponential family in canonical form, it must be writable as

$$f(x \mid \eta) = h(x) \exp\{\eta^\top T(x) - A(\eta)\}$$

with a support (the set of x where $f(x \mid \eta) > 0$) that does not depend on η .

Here the support is $[x_m, \infty)$, which depends on the unknown parameter x_m . Equivalently, the factor $\mathbf{1}_{\{x \geq x_m\}}$ depends on the parameter and cannot be absorbed into $h(x)$ (since h may not depend on parameters). Therefore, when α and x_m are both unknown, the Pareto Type I family is not a (regular) exponential family in canonical form.

If x_m were known (fixed), then

$$f(x \mid \alpha) = \mathbf{1}_{\{x \geq x_m\}} \frac{1}{x} \exp\left\{\alpha(\log x_m - \log x) + \log \alpha\right\},$$

so it can be expressed as a one-parameter exponential family in α (with sufficient statistic $\log x$ up to constants). The obstruction is exactly that the lower cutoff x_m is unknown and changes the support.

2. Assume a logistic regression model (i.e., binary regression under canonical link). Assume that y_i is binary with success probability p^1 and $i = 1, \dots, n$. Specifically, in (simplified) logistic regression, we have that

$$\log\left(\frac{p}{1-p}\right) = X_i^T \beta_{p \times 1},$$

Note that X_i^T is a row vector where its dimension is $1 \times p$. a. Provide the log-likelihood function as a function of the regression parameters. That is, show that

The log-likelihood function for logistic regression is

$$\begin{aligned} \mathcal{L}(p) &= \prod_{i=1}^n f(y_i) \\ &= \prod_{i=1}^n p^{y_i} (1-p)^{1-y_i}, \\ \log \mathcal{L}(p) &= \sum_{i=1}^n [y_i \log(p) + \log(1-p) - y_i \log(1-p)] \\ &= \sum_{i=1}^n \left[y_i \log\left(\frac{p}{1-p}\right) + \log(1-p) \right]. \end{aligned}$$

Recall that $X_i^T \beta_{p \times 1}$ so the log-likelihood becomes

$$\log \mathcal{L}(p) = \sum_{i=1}^n [y_i \mathbf{X}_i^T \beta - \log(1 + \exp(\mathbf{X}_i^T \beta))].$$

In matrix notation, we can write the expression as

$$\log \mathcal{L}(\beta) = \mathbf{y}^T X \beta - \mathbf{1}^T \log(\mathbf{1} + \exp(X \beta))$$

- b. Show (derive) the score functions of the log-likelihood with respect to the regression parameters is as follows:

$$\nabla_{\beta} \log \mathcal{L}(p) = \sum_{i=1}^n \left[y_i \mathbf{X}_i - \frac{\exp(\mathbf{X}_i^T \beta)}{1 + \exp(\mathbf{X}_i^T \beta)} \mathbf{X}_i \right].$$

You may also write the expression in matrix notation as follows:

Define $\pi_i = \frac{\exp(\mathbf{X}_i^T \beta)}{1 + \exp(\mathbf{X}_i^T \beta)}$. Then,

$$\nabla_{\beta} \log \mathcal{L}(p) = X^T (\mathbf{y} - \boldsymbol{\pi})$$

$$\ell(\beta) = \sum_{i=1}^n [y_i X_i^T \beta - \log(1 + \exp(X_i^T \beta))].$$

¹This technically should be p_i , but for simplicity of the derivation, I am suppressing this notation.

Let $\eta_i = X_i^T \beta$. Then

$$\nabla_{\beta} \ell(\beta) = \sum_{i=1}^n [y_i \nabla_{\beta}(\eta_i) - \nabla_{\beta} \log(1 + \exp(\eta_i))].$$

Since $\nabla_{\beta}(\eta_i) = \nabla_{\beta}(X_i^T \beta) = X_i$, and by the chain rule,

$$\nabla_{\beta} \log(1 + \exp(\eta_i)) = \frac{1}{1 + \exp(\eta_i)} \cdot \exp(\eta_i) \cdot \nabla_{\beta}(\eta_i) = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)} X_i,$$

we obtain

$$\nabla_{\beta} \ell(\beta) = \sum_{i=1}^n \left[y_i X_i - \frac{\exp(X_i^T \beta)}{1 + \exp(X_i^T \beta)} X_i \right] = \sum_{i=1}^n (y_i - \pi_i) X_i, \quad \pi_i := \frac{\exp(X_i^T \beta)}{1 + \exp(X_i^T \beta)}.$$

Matrix form Stack X_i^T as rows of X , and let $\pi = (\pi_1, \dots, \pi_n)^T$. Then

$$\sum_{i=1}^n y_i X_i = X^T y, \quad \sum_{i=1}^n \pi_i X_i = X^T \pi,$$

so

$$\nabla_{\beta} \ell(\beta) = X^T (y - \pi).$$

c. Show (derive) the Hessian function of the log-likelihood with respect to the regression parameters is as follows:

$$\nabla_{\beta}^2 \log \mathcal{L}(p) = - \sum_{i=1}^n \left[\frac{1}{1 + \exp(\mathbf{X}_i^T \beta)} \cdot \frac{\exp(\mathbf{X}_i^T \beta)}{1 + \exp(\mathbf{X}_i^T \beta)} \mathbf{X}_i \mathbf{X}_i^T \right].$$

You may also write the expression in matrix notation as follows:

Let $W = \text{diag}\{\pi_1(1 - \pi_1), \dots, \pi_n(1 - \pi_n)\}$. It follows that the Hessian can be written as $-X^T W X$.

$$\nabla_{\beta} \ell(\beta) = \sum_{i=1}^n (y_i - \pi_i) X_i, \quad \pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \quad \eta_i = X_i^T \beta.$$

Differentiate again:

$$\nabla_{\beta}^2 \ell(\beta) = \sum_{i=1}^n \nabla_{\beta}((y_i - \pi_i) X_i) = - \sum_{i=1}^n (\nabla_{\beta} \pi_i) X_i^T,$$

since X_i does not depend on β .

Now compute $\nabla_{\beta} \pi_i$.

Let $\pi_i = g(\eta_i)$ where

$$g(\eta) = \frac{e^{\eta}}{1 + e^{\eta}}.$$

Then

$$g'(\eta) = \frac{e^{\eta}(1 + e^{\eta}) - e^{\eta}e^{\eta}}{(1 + e^{\eta})^2} = \frac{e^{\eta}}{(1 + e^{\eta})^2} = \pi_i(1 - \pi_i).$$

By the chain rule,

$$\nabla_{\beta} \pi_i = g'(\eta_i) \nabla_{\beta} \eta_i = \pi_i(1 - \pi_i) X_i.$$

Substitute back:

$$\nabla_{\beta}^2 \ell(\beta) = - \sum_{i=1}^n \pi_i(1 - \pi_i) X_i X_i^T = - \sum_{i=1}^n \frac{\exp(X_i^T \beta)}{(1 + \exp(X_i^T \beta))^2} X_i X_i^T.$$

In matrix form, define

$$W = \text{diag}(\pi_1(1 - \pi_1), \dots, \pi_n(1 - \pi_n)).$$

Since

$$X^T W X = \sum_{i=1}^n \pi_i(1 - \pi_i) X_i X_i^T,$$

we obtain

$$\nabla_{\beta}^2 \ell(\beta) = -X^T W X.$$

d. Using the previous steps to write out the Newton-Raphson Algorithm in matrix form.

$$\textbf{Newton-Raphson update:} \quad \beta^{(t+1)} = \beta^{(t)} - \left[\nabla_{\beta}^2 \ell(\beta^{(t)}) \right]^{-1} \nabla_{\beta} \ell(\beta^{(t)}).$$

Using

$$\nabla_{\beta} \ell(\beta) = X^T (y - \pi), \quad \nabla_{\beta}^2 \ell(\beta) = -X^T W X,$$

with

$$\pi^{(t)} = \frac{\exp(X \beta^{(t)})}{1 + \exp(X \beta^{(t)})} \quad (\text{elementwise}), \quad W^{(t)} = \text{diag}(\pi_i^{(t)}(1 - \pi_i^{(t)}))_{i=1}^n,$$

we get

$$\beta^{(t+1)} = \beta^{(t)} + \left(X^T W^{(t)} X \right)^{-1} X^T (y - \pi^{(t)}).$$

Equivalently,

$$\beta^{(t+1)}$$

is the solution to the linear system

$$\left(X^T W^{(t)} X \right) \beta^{(t+1)} = \left(X^T W^{(t)} X \right) \beta^{(t)} + X^T (y - \pi^{(t)}).$$

e. A researcher is interested in how variables, such as GRE (Graduate Record Exam scores), GPA (grade point average) and prestige of the undergraduate institution, effect admission into graduate school. The response variable, admit/don't admit, is a binary variable, which can be read in using the command below.

Consider running a logistic regression using the `glm()` function in R below. Write your own function to implement the Newton Raphson algorithm to output the regression coefficients, and verify that the coefficients match those from the `glm()` function.

```
df <- read.csv('https://stats.idre.ucla.edu/stat/data/binary.csv')
mylogit <- glm(admit ~ gre + gpa + rank, data = df, family = "binomial"(link = "logit"))
summary(mylogit)
```

```
##
## Call:
## glm(formula = admit ~ gre + gpa + rank, family = binomial(link = "logit"),
##      data = df)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -3.449548   1.132846  -3.045  0.00233 **
## gre          0.002294   0.001092   2.101  0.03564 *
## gpa          0.777014   0.327484   2.373  0.01766 *
## rank        -0.560031   0.127137  -4.405 1.06e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 499.98  on 399  degrees of freedom
## Residual deviance: 459.44  on 396  degrees of freedom
## AIC: 467.44
##
## Number of Fisher Scoring iterations: 4
```

```
y <- df[,1]
# add column of 1's for the intercept
x <- as.matrix(cbind(rep(1, 400), df[, c(2,3,4)]))

Newton_Raphson_logistic <- function(x, y, b.init, tol = 1e-8, max.iter = 100) {
  b <- as.matrix(b.init)          # ensure column vector
  iter <- 0
  diff <- Inf

  while (diff > tol && iter < max.iter) {
    # linear predictor
    eta <- x %*% b

    # fitted probabilities (logistic)
    p <- exp(eta) / (1 + exp(eta))

    # weights
    w <- as.vector(p * (1 - p))
    W <- diag(w)

    # score and Hessian
    score <- t(x) %*% (y - p)
    H <- -t(x) %*% W %*% x

    # Newton-Raphson update: b_new = b - H^{-1} score
    b.new <- b - solve(H, score)
```

```

    diff <- max(abs(b.new - b))
    b <- b.new
    iter <- iter + 1
  }

  list(coefficients = as.vector(b),
        iterations = iter,
        converged = (iter < max.iter))
}

# run + verify vs glm
b.init <- rep(0, ncol(x))
nr_fit <- Newton_Raphson_logistic(x, y, b.init)

nr_fit$coefficients

## [1] -3.44954840  0.00229396  0.77701357 -0.56003139

coef(mylogit)

## (Intercept)          gre          gpa          rank
## -3.44954840  0.00229396  0.77701357 -0.56003139

```

3. Propose a homework exercise that you believe would help reinforce the class content to students in this class. You may not repeat any exercise explicitly that was used in class and this exercise should be done individually.

Write and propose the question(s), include a rubric, and include a solution.

You may utilize online resource, but please do cite them if you use them and provide details.

Practice Exercise

The geometric distribution is commonly used to model the number of failures before the first success in repeated Bernoulli trials.

Suppose

$$Y_1, \dots, Y_n$$

are independent observations from a geometric distribution with parameter

$$p$$

, where

$$0 < p < 1$$

. The probability mass function is

$$f(y|p) = (1-p)^y p, \quad y = 0, 1, 2, \dots$$

(a)

Rewrite the geometric probability mass function in exponential form. Show that

$$f(y|p) = \exp \{y \log(1-p) + \log p\}.$$

(b)

Show that the geometric distribution belongs to the exponential family.

Write the distribution in the form

$$f(y|\eta) = h(y) \exp\{\eta T(y) - A(\eta)\}.$$

Identify explicitly

- the natural parameter

$$\eta$$

- the sufficient statistic

$$T(y)$$

- the log-partition function

$$A(\eta)$$

- the base measure

$$h(y)$$

(c)

Using your result from part (b), derive the likelihood for

$$n$$

independent observations

$$L(p) = \prod_{i=1}^n f(y_i|p).$$

Write the corresponding log-likelihood and identify the sufficient statistic for

$$p$$

.

(d)

Derive the maximum likelihood estimator (MLE) for

$$p$$

.

(e)

Interpret the sufficient statistic from part (c) in plain English. What information about the sample does it summarize?

Solution

(a)

Start with the probability mass function

$$f(y|p) = (1-p)^y p.$$

Rewrite the expression using logarithms:

$$(1-p)^y p = \exp(y \log(1-p) + \log p).$$

Thus

$$f(y|p) = \exp\{y \log(1-p) + \log p\}.$$

(b)

Recall the exponential family form

$$f(y|\eta) = h(y) \exp\{\eta T(y) - A(\eta)\}.$$

From part (a),

$$f(y|p) = \exp\{y \log(1-p) + \log p\}.$$

Let

$$\eta = \log(1-p).$$

Then

$$1-p = e^\eta$$

and

$$p = 1 - e^\eta.$$

Substituting,

$$f(y|\eta) = \exp\{\eta y + \log(1 - e^\eta)\}.$$

Thus we identify

$$T(y) = y$$

$$\eta = \log(1 - p)$$

$$A(\eta) = -\log(1 - e^\eta)$$

$$h(y) = 1, \quad y = 0, 1, 2, \dots$$

(c)

For independent observations,

$$L(p) = \prod_{i=1}^n (1 - p)^{y_i} p.$$

Simplify:

$$L(p) = p^n (1 - p)^{\sum_{i=1}^n y_i}.$$

The log-likelihood is

$$\ell(p) = n \log p + \left(\sum_{i=1}^n y_i \right) \log(1 - p).$$

Thus the sufficient statistic is

$$T = \sum_{i=1}^n y_i.$$

(d)

Differentiate the log-likelihood:

$$\frac{d\ell}{dp} = \frac{n}{p} - \frac{\sum y_i}{1 - p}.$$

Set equal to zero:

$$\frac{n}{p} = \frac{\sum y_i}{1 - p}.$$

Multiply both sides:

$$n(1 - p) = p \sum y_i.$$

Solve for

$$p$$

:

$$n = p \left(n + \sum y_i \right)$$

Thus the MLE is

$$\hat{p} = \frac{n}{n + \sum_{i=1}^n y_i}.$$

(e)

The sufficient statistic

$$T = \sum_{i=1}^n y_i$$

represents the total number of failures observed before success across all trials.

This means that inference about

$$p$$

depends only on the **total number of failures**, rather than the individual observations.

Grading Rubric (4 Points Total)

Each grading element below is worth 0.25 points. Partial credit should be awarded when the relevant step is clearly demonstrated.

(a) (0.75 points) Rewriting the geometric pmf in exponential form

- Writes the geometric pmf correctly.
- Rewrites powers using logarithms.
- Final expression matches

$$f(y|p) = \exp\{y \log(1 - p) + \log p\}.$$

(b) (1 point) Identifying exponential family components

- Correctly expresses

$$f(y|\eta) = h(y) \exp\{\eta T(y) - A(\eta)\}.$$

- Correctly identifies the natural parameter.
 - Correctly identifies the sufficient statistic.
 - Correctly identifies the log-partition function and base measure.
-

(c) (1 point) Likelihood and sufficient statistic

- Writes the likelihood for independent observations.
- Correctly derives the log-likelihood.
- Collects terms involving

$$\sum y_i$$

- Identifies the sufficient statistic.
-

(d) (1 point) Maximum likelihood estimator

- Differentiates the log-likelihood correctly.
- Sets derivative equal to zero.
- Solves algebraically for

$$p$$

- Correctly identifies the MLE

$$\hat{p} = \frac{n}{n + \sum y_i}.$$

(e) (0.25 points) Interpretation

- Provides a correct interpretation of the sufficient statistic.